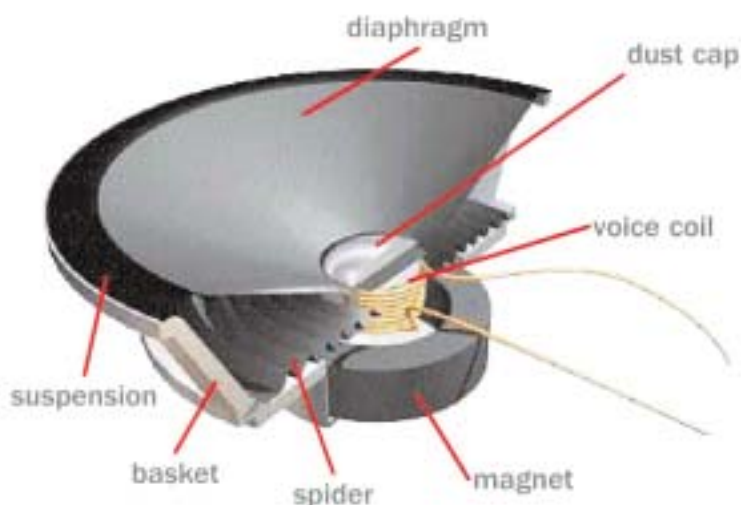


Under the bonnet of a Speaker



Speakers: Do's and Don'ts

Do's:

- ✓ Wipe the speakers using a soft cloth dampened with a mild detergent
- ✓ Lightly vacuum the grille to remove dust particles

Don'ts:

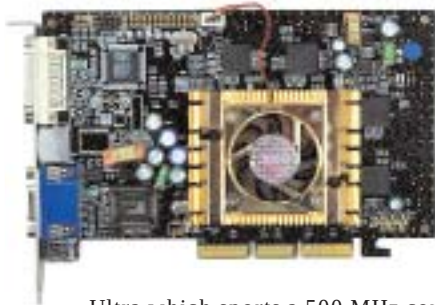
- ✗ Do not listen to music at extreme volumes; reduce the volume when you can notice distortion
- ✗ Never unplug the speakers while they are on
- ✗ If you use an adapter that did not come with your speakers, make sure the ratings of the amplifier (in RMS wattage) and adapter (in ohms) match, or you could cause severe damage to your speakers

The Future of Graphics, Sound and Speakers

The future of graphics cards and gaming has always dominated Web site and magazine headlines. The abiding interest computer buffs have in graphics cards was mainly generated after manufacturers started fiercely competing with each other for the coveted top slot. The gaming industry is the main engine behind this growth. Developers like ID software and EPIC MegaGames continue to release ever-demanding games that require the appropriate hardware.

The reigning king, nVidia, and the almost-there ATi are two

major rivals who keep this arena hot-as-hell by coming out with souped-up technologies as fast as microprocessors change. The latest graphics chipset from nVidia is the GeForce FX 5800



Ultra which sports a 500 MHz core clock and a memory clock of 1 GHz.

The graphics card market last year also saw companies like SiS and Matrox entering the fray with offerings in the mid-range segment, fuelling the competition even more. The largest market anywhere in the world for graphics cards is the value segment, therefore once the company comes out with their most powerful chipset they also start to offer stripped-down versions of the same chipset at much lower, affordable prices.

So what does the future hold for graphics cards? Take two new cards—the Matrox P650 and the P750. These cards are equipped with some improved technologies like dual DVI output, dual-display gamma correction, fully symmet-

ric RGB output, Matrox-dual-display plus TV output and older triple-head output. And ATi and nVidia are working hard to come out with better and improved drivers to eliminate any issues their cards may currently have with some games and operating systems.

Trident and VIA (Savage graphic chip) are also in the race but in a very small way. Trident offers a very low-level solution (Blade XP) and VIA integrates the Savage Pro graphics chip into their motherboards because there's virtually no market for their graphic solutions if sold separately. But not everyone requires the best graphics cards, and not many can afford them. So most of these companies will survive some way or the other.

Speakers and sound cards don't change at the same pace as graphics cards. However, players like Creative (whose products are widely available in India), Zoltrix and Turtle Beach are offering sound cards with improved technology, but these cards are mainly bought by professionals rather than regular desktop users because of the high prices. American company M-Audio has also started offering a 7.1-channel surround sound card, increasing the number of speakers to eight, with 24-bit/192 KHz audio for enhanced audio experience.

Some labs are already working on 9.1-channel surround sound. This increase in the number of channels (and hence, speakers) will not only make future movie experiences more immersive, but also give hardcore gamers an edge over rivals by allowing them to precisely predict where an attack is coming from, and move out of harm's way.

There are more models of speakers available now than ever before, and there's been tremendous improve-

ment in packaging as well. Harmon Cardon speakers, for instance, come with a glass shell in which the drivers are enclosed. Before long, speakers could well become a design statement.

